

Dr. Gerald Laura-Moore

Doctorate in Neurotechnology | Data Scientist | Machine Learning Engineer | Developer | Researcher

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Profile

Senior Machine Learning (ML) Engineer with expertise encompassing computer vision, time-series analysis, agentic AI systems, and production ML systems. High impact experience in developing ML solutions for a variety of tasks and scaling from research to production systems. Domain expertise in geospatial remote sensing, neuroscience and astrophysics. Passionate about building innovative solutions and expanding my knowledge in an environment that encourages learning and discovery.

Experience

Dec 2022 - present

Senior Machine Learning Engineer | [Agreena, Copenhagen, DK](#)

- Senior technical lead for DS, ML engineering and software engineering using remote sensing for verification in a soil carbon program.
- Led technical development across numerous projects: agricultural land detection (FracTAL ResUNet), crop detection and analysis (Mask R-CNN), time-series event detection and prediction (1D CNN, LSTM, CatBoost, XGBoost, Transformers), satellite scene classification (2D CNN, MobileNet, EfficientNet), global-scale event mapping (LSTM, Transformers), image report generation (VLM), heterogeneous treatment effects through causal ML analysis (Causal Forest) and agentic workflows (Pydantic AI, LangChain, LangGraph, FastMCP).
- Solo development of a model to distill embeddings from a geospatial foundational image model to improve, optimise and standardise internal ML pipelines.
- Developed image report generation with multi-task classification for field event verification (fine-tuning PaliGemma, LLaVA, Gemma 3) and ground truth data collection using multi-task classifiers on Google Street View imagery.
- Built user-facing agentic AI chatbot leveraging custom MCPs for unique data-driven insights.
- Scaled research-to-production pipelines using Ray, FastAPI, AnyScale, VertexAI, Airflow and Kubernetes, managing KPIs for accuracy, throughput, resource utilisation and stakeholder satisfaction.
- Recognised by CEO and senior leadership for driving value and delivering results with significant contribution to company direction.
- Led project planning and team coordination to meet KPI deliverables whilst advancing state-of-the-art research and tooling adoption.
- Developed Python SDK and data ingestion service for STAC catalogues and API access, driving efficiency across DS and Engineering teams.
- Implemented PCA and manifold-learning for data drift detection and K-fold cross-validation for dataset split optimisation.
- Addressed dataset limitations through ensemble learning, auxiliary task learning and transfer learning.
- Authored publications, presented at conferences and delivered technical presentations to stakeholders and prospects.
- Led hiring and served as go-to-trustee within the team to help address complex challenges.

Jan 2021 - Dec 2022

Data Scientist | [Hummingbird Technologies, London, UK](#)

- Nurtured key client partnerships by researching and developing new customer driven solutions, including instance segmentation approach for high resolution object detection in agricultural fields, and an attention based semantic segmentation model for delineating agricultural parcels at scale.
- Led grant-funded research with documentation for research awards and conference presentations.

Oct 2016 - Jan 2021

Ph.D Neurotechnology | [Imperial College London, London, UK](#)

- Sole ownership and development of an end-to-end neurone analysis pipeline for multi-terabyte two-photon microscopy data, including CNN classification through custom or adapted architectures (VGG Net, Google Inception), semantic segmentation (UNet, UNet++), image registration (Elastix, aMAP, ANTs) and dashboard visualisations (Plotly, Dash, Heroku).
- Applied the pipeline to investigate neuronal connectivity in the visual thalamic and pre-frontal cortex systems, as well as studying structural changes under Alzheimers and Huntingtons pathology.
- Presented research through publications, conferences, and workshops.

Projects

[TinyEarth](#)

- A lightweight distilled model adaptation to recreate Googles AlphaEarth embeddings (see [report](#)).
- Dual-stream encoder fusing Sentinel-1 & 2 with a temporal attention mechanism, leveraging von Mises-Fisher loss for embedding reconstruction.

[BinOcular](#)

- Self driven development of a Python package that clusters camera photographs based on feature similarity and temporal thresholding.
- Uses a pre-trained EfficientNet model to compress imagery into a latent feature space, and subsequently uses a cosine-similarity metric and temporal thresholding to group images into *similar* clusters based on shared features within images.

Education

2015 - 2016 | MRes Neurotechnology | [Imperial College London, UK](#)

2011 - 2015 | MPhys in Physics and Astrophysics | [University of Sussex, UK](#)

2009 - 2011 | A-levels including Mathematics, Physics and Computing | [Furze Platt Sixth Form College, UK](#)

2007 - 2009 | Thirteen GCSEs including Mathematics, Physics and Biology | [Furze Platt Senior School, UK](#)

Technical Skills

Languages | Python, C++, JAVA, MATLAB, Rust

Libraries | PyTorch, Tensorflow, Keras, Ray, Scikit-learn, Scipy, OpenCV, Pydantic, Pydantic AI, LangChain/Graph, Pytest

Databases | STAC, MySQL, PostgreSQL

Software | AnyScale, Docker, Git, CircleCI, Poetry, GCP, Vertex AI, ImageJ/Fiji, Blender, Adobe Suite

Publications

2025

Longitudinal testing of exploratory behaviour in mice reveals stable cognitive traits across the adult lifespan | Rushdie Abuhamdah, Gerald Moore, Deyl Djama, Florian Zirpel, Chris Edge, Abdel Ennaceur, Paul Chazot, Diana Cash, Eugene Kim, Anthony Vernon, Paul Chadderton, Stephen Brickley. *Aging Cell*. [10.1111/accel.70287](#).

Hierarchical Bayesian modeling of multi-region brain cell count data | Sydney Dimmock, Benjamin MS Exley, Gerald Moore, Lucy Menage, Alessio Delogu, Simon R Schultz, E Clea Warburton, Conor Houghton, Cian O'Donnell. *eLife Neuroscience*. [10.7554/eLife.102391.1](#).

2024

Detecting cover crop activity at scale using fusion of multiple satellite sources | Gerald Moore, Edward Dowling, Gabor Szakacs, Daniel Szponar. *Remote Sensing* [to be submitted].

Tracking tillage practices across Europe using multi-source Earth observations & machine learning | Nicholas Synes, Aoife Whelan, Edward Dowling, Francois Lemarchand, Khushboo Jain, Ben Smith, Gerald Moore, Peter Kongstad, Blayne Lees, Vincent Cornwell, Nathan Torbick. *IEEE IGARSS*.

The type of inhibition provided by thalamic interneurons alters the input selectivity of thalamocortical neurons | Stephen Brickley, Deyl Djama, Florian Zirpel, Zhiwen Ye, Gerald Moore, Charmaine Chue, Christopher Edge, Polona Jager, Alessio Delogu. *bioRxiv*. [10.1016/j.crneur.2024.100130](#).

2021

Dual midbrain and forebrain origins of thalamic inhibitory interneurons | Polona Jager, Gerald Moore, Padraic Calpin, Xhuljana Durmishi, Yoshiaki Kita, Irene Salgarella, Yan Wang, Simon R. Schultz, Stephen Brickley, Tomomi Shimogori, Alessio Delogu. *Elife*. [10.7554/eLife.59272](#).

2020

Cell counting in targeted nuclei of whole brain two-photon image data | Gerald Moore, Polona Jager, Alessio Delogu, Simon Schultz, Stephen Brickley. *Biophotonics Congress: Biomedical Optics Congress 2018 (Microscopy/Translational/Brain/OTS)*, OSA Technical Digest (Optical Society of America, 2018). [10.1364/TRANSLATIONAL.2018.JTu3A.48](#).

Conferences & Workshops

2022

Living Planet Symposium | Presented FracTAL ResUNet for field boundary detection and Mask R-CNN for crop counting in drone imagery.

2019

British Neuroscience Association | Presented U-Net based cell counting and distribution analysis method for whole mouse brain microscopy data. Presented longitudinal research of cognitive decline in female mice and its association with brain ageing.

London Neurotechnology Network Imaging Workshop | Presented high-resolution imaging research for mapping small-scale objects across large tissue volumes.

2018

The Optical Society Annual Meeting | Presented an ML approach for cell counting in targeted nuclei of whole brain two-photon microscopy data.

Dementia Symposium ICL Alzheimers Research | Showcased research on brain pathology in response to Alzheimers and Huntingtons disease.

2017

Tissue Clearing and 3D Imaging Course | Learning tissue clearing techniques and light-sheet microscopy.

British Neuroscience Association | Presented macroscopic imaging of neuronal connectivity related to health and disease, as well as meso- and micro-scale changes in synaptic connectivity with age.